Theory Note 3 of 8

Capture Filters vs Display Filters in Wireshark

Network Traffic Analysis, Nmap and Windows Logging | Week 2

1. Overview

Wireshark has two separate filtering systems, and mixing them up is a common beginner mistake. A capture filter decides which packets are saved in the first place. A display filter only changes which of the already captured packets are shown. The two use different syntax, so a filter that works in one box will often be rejected in the other.

2. Key differences

Point	Capture filter	Display filter
When it is applied	Before capture starts. Packets that do not match are never recorded.	After capture. It only hides packets that are already in the file.
Syntax	BPF (Berkeley Packet Filter), for example: host 10.0.0.5 and port 80	Wireshark field syntax, for example: ip.addr == 10.0.0.5 && tcp.port == 80
Can it be changed later?	No. The capture must be restarted.	Yes, at any time, without losing data.
Protocol awareness	Basic: hosts, networks, ports and a few protocols.	Detailed: thousands of fields such as http.host, dns.qry.name or tls.handshake.
Best used for	Long or busy captures, limited disk space, or focusing on one host.	Analysis, hunting and investigation of a saved capture.
Where to enter it	Capture > Capture Filters, or the filter box beside the interface.	The filter bar at the top of the main window.

3. Ten capture filter examples (BPF)

No.	Filter	What it does
1	`host 192.168.1.10`	Traffic to or from one host.
2	`src host 192.168.1.10`	Only packets sent by that host.
3	`dst host 192.168.1.20`	Only packets going to that host.
4	`net 192.168.1.0/24`	Everything inside the subnet.
5	`port 80`	Traffic on port 80 in either direction.
6	`tcp port 443`	TCP traffic on 443 (HTTPS) only.
7	`udp port 53`	DNS over UDP.
8	`icmp`	Only ICMP packets, such as ping.
9	`not arp and not broadcast`	Removes ARP and broadcast noise.
10	`tcp[tcpflags] & tcp-syn != 0 and tcp[tcpflags] & tcp-ack == 0`	Only initial SYN packets, useful for spotting scans.

4. Ten display filter examples

No.	Filter	What it does
1	`ip.addr == 192.168.1.10`	Any packet where that IP is the source or destination.
2	`tcp.port == 22`	TCP traffic on port 22 (SSH).
3	`http.request.method == "GET"`	All HTTP GET requests.
4	`dns.qry.name contains "example"`	DNS queries whose name contains the word example.
5	`icmp.type == 8`	ICMP echo requests (pings).
6	`tcp.flags.syn==1 && tcp.flags.ack==0`	Initial SYN packets only (start of a handshake or a SYN scan).
7	`tcp.flags.reset==1`	Every TCP reset in the capture.
8	`tls.handshake.type == 1`	TLS Client Hello packets, which reveal the server name.
9	`http.response.code >= 400`	HTTP client and server errors.
10	`!(arp || icmp || dns)`	Hides ARP, ICMP and DNS to cut the noise.

Key takeaway
Use a capture filter when you must limit what is recorded, and a display filter when you want to explore what you already have. If you are unsure which to use, capture everything and filter on display, because you cannot recover packets that were never captured.